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Successful bacteriophage treatment of infection involving cardiac implantable electronic device and aortic graft: a Trojan horse concept

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We discuss the case of a 41-year-old man with a right-sided pocket infection and a history of extensive aortic surgery as well as a right carotid subclavian bypass in the presence of Marfan syndrome. During explantation of the cardiac resynchronization therapy with defibrillator system a fistula to the right carotid subclavian bypass surrounded with pus was found. A ¹⁸F-fluorodeoxyglucose positron-emission tomography computed tomography was performed and a generalized vascular graft infection was revealed (Panel A). A vacuum assisted closure treatment was initiated combined with antibiotics. Since no infection control was achieved, local application of a bacteriophage therapy was added. Follow-up showed good infection control and reduced glucose uptake in positron-emission tomography—computed tomography (Panel B).

Such an infection might be complicated by the formation of a biofilm, which is an important source for antibiotic tolerance and treatment failure. Bacteriophages can specifically kill free-floating bacteria but also degrade matrix and penetrate biofilms, targeting antibiotic-tolerant persistent cells, characterized by a slow- or non-growing state. Phages are self-propagating organisms, which regulate themselves at the site of infection, are associated with low risks of adverse events and can stop replication when the target is destroyed. Bacteriophage therapy may be the Trojan horse by which therapeutic access to complex cardiovascular device infections can be gained.

The full-length version of this report can be viewed at: <https://www.escardio.org/Education/E-Learning/Clinical-cases/Electrophysiology>.

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